

* ESPERANZA MATEMÁTICA

$$\left\{ E(x) = \sum_{i=1}^n x_i \cdot f(x_i) \right\} \text{ V. DISCRETA } \quad \overline{P(x=x_i) = f(x_i)}$$

Cuando las probabilidades son iguales $\left\{ E(x) = \frac{x_1 + x_2 + \dots + x_n}{n} = \mu \right\}$
 MEDIA

$$\left\{ E(x) = \int_{-\infty}^{\infty} x \cdot f(x) \cdot dx \right\} \text{ V. CONTÍNUA}$$

PROPIEDADES

$$a) \left\{ \begin{array}{l} \bullet E(Q(x)) = \sum_{i=1}^n Q(x_i) \cdot f(x_i) \quad \text{V. DISCRETA} \\ \bullet E(Q(x)) = \int_{-\infty}^{\infty} Q(x) \cdot f(x) \cdot dx \quad \text{V. CONTÍNUA} \end{array} \right.$$

$$b) \left\{ \begin{array}{l} \bullet E(k) = k \quad k = \text{cte.} \\ \bullet E(kx) = k \cdot E(x) \\ \bullet E(k+x) = k + E(x) \end{array} \right.$$

$$c) \left\{ \begin{array}{l} \bullet E(x+y) = E(x) + E(y) \\ \bullet E(x \cdot y) = E(x) \cdot E(y) \end{array} \right.$$

Sean x e y independientes o no

(Si hubiera mas de dos variables es lo mismo.)

Para dos variables x e y con función de distribución conjunta:

$$f(x, y)$$

$$E(x, y) = \sum_{x_i=1}^{x_i=n_x} \sum_{y_i=1}^{y_i=n_y} x_i y_i f(x_i, y_i) \quad \text{V. DISCRETA}$$

$$E(x, y) = \int_x \int_y x y f(x, y) \cdot dx \cdot dy \quad \text{V. CONTINUA}$$

* MOMENTOS

* MOMENTOS RESPECTO A LA MEDIA.

$$\mu_r = E[(X - \mu)^r] \quad r = 1, 2, 3, \dots$$

$$\mu_0 = 1 \quad \mu_0 = E[(X - \mu)^0] = E[1] = 1$$

$$\mu_1 = 0 \quad \mu_1 = E[(X - \mu)^1] = E(X) - \mu = \mu - \mu = 0$$

$$\mu_2 = \text{Var}(X) \quad \mu_2 = E[(X - \mu)^2] = E(X^2) - \mu^2 = \text{Var}(X)$$

En forma general: $\mu_r = \sum_{i=1}^n (x_i - \mu)^r \cdot f(x_i)$ V. DISCRETA

$\mu_r = \int_{-\infty}^{\infty} (x - \mu)^r \cdot f(x) \cdot dx$ V. CONTINUA

* MOMENTO RESPECTO AL ORIGEN

$$\mu'_r = E[x^r] \quad r = 1, 2, 3, \dots$$

$$\mu'_0 = 1 \quad \mu'_0 = E[x^0] = E[1] = 1$$

$$\mu'_1 = \mu \quad \mu'_1 = E[x^1] = \mu$$

En forma general: $\mu'_r = \sum_{i=1}^n x_i^r \cdot f(x_i)$ V. DISCRETA

$\mu'_r = \int_{-\infty}^{\infty} x^r \cdot f(x) \cdot dx$ V. CONTINUA

* RELACIÓN ENTRE MOMENTOS

$$\mu_r = \mu_r' \text{ Completar}$$

Analizando los cinco primeros momentos :

$$\mu_0 = 1$$

$$\mu_1 = 0$$

$$\mu_2 = \mu_2' - \mu^2$$

$$\mu_3 = \mu_3' - 3\mu_2'\mu + 2\mu^3$$

$$\mu_4 = \mu_4' - 4\mu_3'\mu + 6\mu_2'\mu^2 - 3\mu^4$$

FUNCIÓN GENERATRIZ DE MOMENTOS

$$M_X(t) = E(e^{tx})$$

$$M_X(t) = \sum_{i=1}^n e^{tx_i} f(x_i)$$

V. DISCRETA

$$M_X(t) = \int_{-\infty}^{\infty} e^{tx} f(x) dx$$

V. CONTÍNUA

$$\mu_r' = \left. \frac{d^r M_X(t)}{dt^r} \right|_{t=0}$$

* VARIANZA Y DESVIACIÓN TÍPICA.

$$\text{Var}(x) = \sum_{i=1}^n [x_i - E(x)]^2 \cdot f(x_i)$$

σ

V. DISCRETA

$$\text{Var}(x) = E[(X - \mu)^2] = \sum_{i=1}^n [x_i - \mu]^2 \cdot f(x_i)$$

mayúscula

$$\text{Var}(x) = \int_{-\infty}^{\infty} [x - E(x)]^2 \cdot f(x) \cdot dx$$

σ

V. CONTÍNUA

$$\text{Var}(x) = E[(X - \mu)^2] = \int_{-\infty}^{\infty} (x - \mu)^2 \cdot f(x) \cdot dx$$

mayúscula

$$\text{Var}(x) = \sigma^2$$

PROPIEDADES

Siendo $\mu = E(x)$

$$a) \text{Var}(x) = E[(X - \mu)^2] = E(x^2) - \mu^2$$

$$\begin{aligned} \text{Var}(x) &= E[(X - \mu)^2] = E(x^2 - 2\mu x + \mu^2) \\ &= E(x^2) - 2\mu E(x) + \mu^2 \\ &= E(x^2) - 2\mu^2 + \mu^2 \\ &= E(x^2) - \mu^2 \end{aligned}$$

$$b) \text{Var}(k) = 0$$

$$\text{Var}(x) = E[(k - E(k))^2] = E[(k - k)^2] = E(0) = 0$$

$$\begin{aligned}
 c) \quad \boxed{\text{Var}(X+k) = \text{Var}(X)} \quad & \text{Var}(X+k) = E\{[(X+k) - E(X+k)]^2\} \\
 & = E\{(X+k)^2 - 2(X+k)E(X+k) + (E(X+k))^2\} \\
 & = E\{(X+k)^2 - 2(X+k)(E(X)+k) + (E(X)+k)^2\} \\
 \text{Var}(X+k) & = E\{X^2 + 2kX + k^2 - 2XE(X) - 2Xk - 2kE(X) - 2k^2 + E(X)^2 + 2E(X)k + k^2\} \\
 & = E\{X^2 - 2X\mu + \mu^2\} \\
 \text{Var}(X+k) & = E(X^2) - 2\mu E(X) + \mu^2 = E(X^2) - 2\mu^2 + \mu^2 = E(X^2) - \mu^2 = \text{Var}(X)
 \end{aligned}$$

$$\begin{aligned}
 \boxed{\text{Var}(k \cdot X) = k^2 \cdot \text{Var}(X)} \quad & \text{Var}(kX) = E\{[kX - E(kX)]^2\} \\
 & = E\{(kX)^2 - 2(kX)E(kX) + (E(kX))^2\} \\
 & = E\{k^2X^2 - 2k^2XE(X) + k^2E(X)^2\} \\
 & = k^2E(X^2) - 2k^2E(X)E(X) + k^2E(X)^2 \\
 & = k^2E(X^2) - 2k^2\mu^2 + k^2\mu^2 \\
 & = k^2E(X^2) - k^2\mu^2 \\
 & = k^2(E(X^2) - \mu^2) \\
 & \quad \text{Var}(X) \\
 \text{Var}(kX) & = k^2 \text{Var}(X)
 \end{aligned}$$

X e Y son dos variables aleatorias INDEPENDIENTES :

$$\boxed{\text{Var}(X+Y) = \text{Var}(X) + \text{Var}(Y)}$$

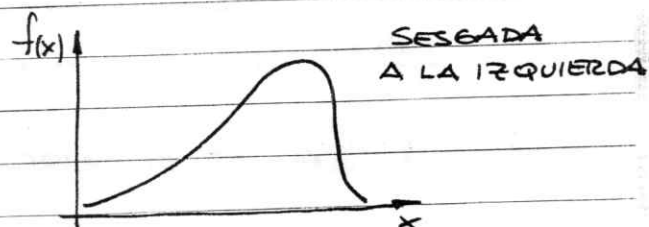
$$\boxed{\text{Var}(X-Y) = \text{Var}(X) + \text{Var}(Y)}$$

$$\boxed{\text{Var}(X \cdot Y) = \text{Var}(X) \cdot \text{Var}(Y)}$$

* OTRAS MEDIDAS DE DISPERSIÓN

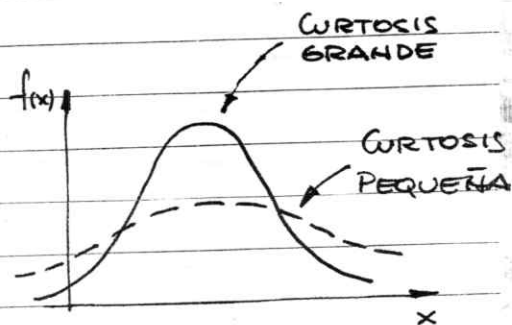
* SESGO

$$\alpha_3 = \frac{E[(X-\mu)^3]}{\sigma^3} = \frac{\mu_3}{\sigma^3}$$



* CURTOSIS

$$\alpha_4 = \frac{E[(X-\mu)^4]}{\sigma^4} = \frac{\mu_4}{\sigma^4}$$



* DESVIACIÓN MEDIA

$$DM(x) = E[|X-\mu|] = \sum_{i=1}^n |x_i - \mu| \cdot f(x_i) \quad \text{V. DISCRETA}$$

$$DM(x) = E[|X-\mu|] = \int_{-\infty}^{\infty} |x - \mu| f(x) \cdot dx \quad \text{V. CONTINUA}$$

$$* \text{RECORRIDO INTERCUARTÍLICO} = x_{75} - x_{25}$$

$$* \text{RECORRIDO SEMI-INTERCUARTÍLICO} = \frac{x_{75} - x_{25}}{2}$$

$$* \text{RECORRIDO} = x_{\max} - x_{\min} \quad (\text{No tiene sentido si algunos de los extremos es } \infty)$$